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A Study on Driver State Recognition Using CNN-based Multimodal Multi-input Learning*

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Abstract

Accurate identification of a driver's state is extremely important for the safety of both the driver and passengers. To date, many studies have demonstrated that identifying a driver's state by creating an image of their face and upper body using a camera and then learning these CNN-based neural networks can be effective to a certain extent. In addition, efforts have been made to identify a driver's level of fatigue using electroencephalogram (EEG), heart rate (ECG), and electrooculogram (EOG) data measurements, multiple sensors, and the RNN-LSTM model method. At present, increasing accuracy can be achieved using simple sensor devices and AI structures that can be installed in small devices such as on-device AI. In this study, we used a driver's facial expression and upper body along with the sound data generated during driving to recognize the driver's state more accurately. Subsequent experiments revealed that CNN-based neural network learning alone, using triple input elements, improved accuracy. To apply on-device AI, we proposed a CNN with a simple structure that can collect data employing only a camera and a recorder. We compared the proposed method with learning in ResNet50 and Xception, which revealed that it works effectively. These experimental results indicate that CNN can be used in multimodal applications and can be an efficient choice over other complex neural network learning methods that utilize multimodal learning data.

Keywords : Artificial Intelligence, Multimodal, Multi Input, On-device AI, CNN, Driver State Recognition

Major Classification Code : Computers(L63), Computer Software (L86), Technological Change (O33)

1. Introduction

Understanding a driver's emotions and state while driving is extremely important for the safety of both the driver and passengers, and multiple studies have been

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conducted to determine how optimally this can be achieved. Researchers are currently proposing a system that automatically warns of danger through AI learning by using all driver-centered data, such as their face, body signals, driver appearance, gaze, voice and surrounding sounds, as information for understanding their emotional state. The driver's facial information is considered the best information for judging their emotions. Lee et al. (2018) proposed a CNN-based emotion classification method that distinguishes between aggressive and smooth driving using the driver's facial image. To test this, they detected the driver's emotions using near-infrared (NIR) and thermal imaging camera sensors and achieved 99% accuracy. Zaman et al. (2022) sought to recognize seven emotions using a driver's facial image. They used an improved Faster R-CNN and a neural architecture search network (NASNet), achieving a recognition accuracy of 99.15%. An attempt was also made to identify the driver's arousal state using measurements of body signals, rather than facial expressions. Nakisa et al. (2020) used electroencephalography (EEG) and blood volume pulse (BVP) measures as CNN-LSTM learning data. They tested four driver states: high-arousal-positive emotions, low-arousal-positive emotions, high-arousal-negative emotions, and low-arousal-negative emotions, and reported an accuracy of 71%. Numerous studies have also demonstrated that the ability to learn a driver's appearance while driving could help predict safe driving by identifying distracted behavior. This is because drivers' increased use of digital devices such as mobile phones can decrease safety, regardless of their emotions. For instance, Ye et al. (2020) detected a driver's distracted behavior using a channel attention CNN method. This involved using an SE module that assigns different weights according to importance. A subsequent experiment employing the SE-Xception network yielded an accuracy of 92.6%. Lyu et al. (2020) analyzed driver behavior by learning driver gaze and head pose information using CNN, which yielded an accuracy of 81%. Han and Cho (2021) trained 10 classes of data using CNN and improved ResNet-101 and achieved an F1 Score of 98.00%: the data classes were Phone Use, Smoking, Music Adjustment, Drinking, Loss of Control, Looking Away, Navigation, Pet Care, Voice Call, and Smart Device Manipulation. Xu et al. (2022) reported an accuracy of 84% utilizing an aggressive driving behavior prediction method comprising a Hidden Markov Model (HMM) and a Long Short-Term Memory (LSTM), while Park et al. (2024) proposed using six classes of image data (Finding an item, Drunk driving, Drowsy driving, Vehicle control, Phone call, Mobile phone handling) using a Vision Transformer-based model to detect driver abnormal behavior, achieving an accuracy of 70%-96% for each classification item.

Alongside these studies, the results of studies investigating human emotion recognition using voice information have also been reported and show high accuracy. For instance, Choee et al. (2019) used CNN-based transfer learning with English and German voices and a fine-tuning method to distinguish four emotions with 95% accuracy. Nam (2023) proposed DnCNN for voice emotion recognition exposed to noise. The average recognition rate of a voice without noise that distinguished five emotions (neutral, happy, sad, angry, and fear) was 94.3%. When noise was added, DnCNN achieved high recognition rates of 90.7% for subway transfer stations, 89.4% for cafeterias, and 90.4% for subways. Kim and Kwon (2023) developed an emotion recognition model employing a multimodal method that simultaneously uses voice and image data. Voice data were represented as Mel Spectrogram, and image data were extracted through a Slowfast feature extractor. The results revealed that the model using 2D-CNN classified seven emotions and exhibited 60.11% accuracy. In this paper, we propose a multimodal driver status recognition system that utilizes images of the face, driving posture, and sound information, including the driver's voice. Given the increasing interest in on-device AI, we tested whether it would be possible to distinguish the driver's status through images and sound information that can be measured with a regular mobile phone using a simple CNN structure. Experiments revealed that the system was 99% accurate. The contributions of this paper are as follows.

(1) We proposed a multimodal CNN that learns face and driving posture images, the driver's voice, and surrounding sound data to distinguish the driver's status. The experimental results exhibited 99.9% accuracy.
(2) We proposed a CNN system that distinguishes six conditions in total, including three driver conditions (normal, fainting, and drowsy) and three driver emotions (surprise, anger, and anxiety) that can cause abnormal driver states.
(3) Taking into consideration on-device AI, only images and sounds that can be measured with simple digital instruments were used. This revealed that good results can be achieved even with simple CNN learning.

2. Related Works

Recent studies on driver status recognition for safe driving have focused on confirming various driver situations through images, determining fatigue or stress levels by measuring body signals, and distinguishing between drowsy and non-drowsy drivers by analyzing the shape of their eyes and lips. Each method yielded good results with an accuracy

of 95%-99%. In this study, we proposed a system that can distinguish three types of driver states: normal, faint, and drowsy, and three types of driver emotions that could cause abnormal driver states: surprise, anger, and anxiety. For this research, we combined existing studies on recognizing driver states and emotions into one system. The results are expected to be more useful in real-life situations than existing studies.

Table 1: Examples of drivers' distracted behaviors

Drivers' distracted behaviors	
Our six classifications	c0: normal c1: faint c2: drowsy c3: surprise c4: anger c5: anxiety
Ten classifications	c0: safe driving c1: texting on the phone - right hand c2: talking on the phone - right hand c3: texting on the phone - left hand c4: talking on the phone - left hand c5: operating the radio c6: drinking c7: reaching behind c8: hair and makeup c9: talking to passenger

Table 1 presents examples of drivers' distracted behaviors used by ourselves and other researchers.

Among recent papers proposing driver state recognition models, those that studied drowsy driving classification have reported promising results. Kim et al. (2024) proposed a drowsiness recognition system based on SERN (Squeeze and Excitation Resnet Network) to recognize driver drowsiness. It learned with 99.84% accuracy, mainly from image data of the eyes and lips on the face. Das et al. (2024) used the CNN-LSTM algorithm with video clip and IoT sensor data to distinguish a driver's drowsy or non-drowsy state with 98.8% accuracy. In addition, other papers have captured the shape of the eyes and blinking on the driver's face and also reported promising results. Madni et al. (2024) used CNN with the VGG16-LGBM method to capture image data of eye movement on the face and expressed the driver's drowsiness state with 99% accuracy. Ahmed et al. (2022) performed CNN-based learning, such as MTCNN and InceptionV3, using image data of the eyes and lips on the face and classified the drowsy driving state with 97.1% accuracy. Abbas and Alsheddy (2021) also employed a method that combined CNN and RNN-LSTM models. They collected data using eye-blink images, along with electroencephalogram (EEG), heart rate (ECG) and electrooculogram (EOG) signals, to measure driver fatigue, which can cause drowsy driving. Their method yielded 94.5% accuracy.

Table 2: Summary of significant recent achievements and the research methods employed to investigate driver state

Papers	Number of class	Classification	Data	Multi-modal	Methods	Accuracy (%)
Ours	6	Six: Drivers' distracted Behaviors	Driver image, sound spectrogram	O	multi-input CNN	99.9
Kim et al. (2024)	2	Two: Drowsy, Normal	Face image (eye, mouth)	X	CNN(SERN)	99.84
Das et al. (2024)	2	Two: Drowsy, Normal	Driver image, signal (IoT sensors)	O	CNN-LSTM	98.8
Madni et al. (2024)	2	Two: Drowsy, Normal	Face image (eye movement)	X	CNN (VGG16-LGBM)	99
Ahmed et al. (2022)	2	Two: Drowsy, Normal	Face image (eye, mouth)	X	MTCNN, InceptionV3	97.1
Abbas & Alsheddy (2021)	2	Two: Fatigue, Normal	Eye-blink, signal (EEG,ECG,EOG)	O	CNN RNN-LSTM	94.5
Amin et al. (2023)	3	Three: Stress levels (low, medium, high)	Driver image, signal (ECG, HR, etc.)	O	CNN-LSTM	96.6
Mou et al. (2021)	3	Three: Stress levels (low, medium, high)	Driver image, Signal (eye, vehicle data, environmental data)	O	CNN-LSTM	95.5
Doniec et al. (2020)	4	Four: Driving activity (parking, roundabout, city traffic, intersection)	Signal (EOG)	X	1D CNN	95.6
Qin et al. (2022)	10	Ten: Drivers' distracted Behaviors	Driver image (HOG feature)	X	D-HCNN	99.87
Huang et al. (2020)	10	Ten: Drivers' distracted Behaviors	Driver image	X	CNN(HCF)	96.74

Other papers have studied driver stress levels using body signals. For instance, Amin et al. (2023) classified driver stress levels into three categories (low, medium, and high) and used driver images and body signals such as ECG, HR, GSR, EMG, and RESP. Learning was performed by mixing CNN and LSTM methods, and stress levels were distinguished with 96.6% accuracy. Mou et al. (2021) used CNN and LSTM methods, along with images of the driver's eye condition and car and external environment data, to distinguish three stress levels with 95.5% accuracy. Doniec et al. (2020) used the driver's signal (EOG) information during actual driving to identify four driving activities (parking, roundabout, city traffic, and intersection). They then used CNN to identify the driving status with 95.6% accuracy. Some papers have distinguished drivers' distracted behaviors. For example, Qin et al. (2022) implemented a small-sized D-HCNN using HOG (Histogram of Oriented Gradient) features extracted from the driver's image and distinguished 10 driver states with 99.87% accuracy. Huang et al. (2020) employed a hybrid CNN using the driver's image and achieved 96.74% accuracy.

The aim of this study was to identify the driver's state more realistically by adding an emergency, such as fainting, to the simple distinction between drowsy and normal states previously presented. In addition, a model was proposed that distinguishes six driver states by discerning emotions such as surprise, anxiety, and anger, which can cause abnormal driving conditions, through the driver's facial expressions. In addition to images of the driver's state, the data used were expressed as sound data in spectrogram form. Taking into consideration the implementation of on-device AI, a simple CNN structure capable of multi-input in a multimodal manner was constructed, and 99.9% accuracy was achieved. Table 2 summarizes significant recent achievements, along with the research methods employed, and compares them with the characteristics of the current study.

3. Proposed System

We proposed a multimodal driver state recognition system, including a multi-input CNN that receives three image data inputs, learns through convolution layers, and goes through a concatenation process. As a factor considered in preparing the data, we collected images and sounds by utilizing cameras and sound recorders commonly installed in digital devices such as mobile phones, considering the trend of on-device AI. The driver's voice and environmental sound were imaged as spectrograms from sound wave files. The driver-face, driver-state, and sound spectrogram images were the three input data for learning and inference.

The proposed multi-input CNN structurally receives three image data, performs concatenation after two convolution layers, undergoes a learning process with three convolution layers and three affine layers, and outputs six classes through the final softmax layer. This represents a modified structure of the multi-input CNN proposed by Han (2024) which was based on Alexnet. Figure 1 presents the overall structure of the proposed multimodal driver state recognition system.

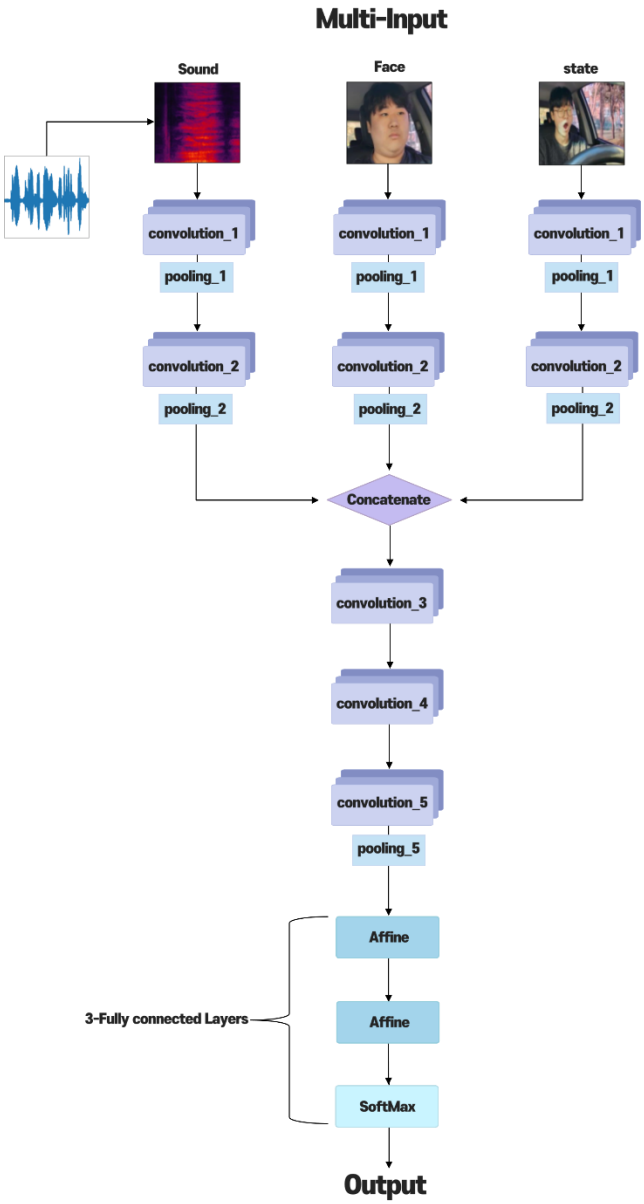


Figure 1: Overall structure of the proposed system.

Table 3 lists the detailed specifications of the eight-layer multi-input CNN, including the parameters used for learning and inference.

Table 3: Detailed specifications of the multi-input CNN

Layer	Input 1	Input 2	Input 3
Input pixel size	227*227*3	227*227*3	227*227*3
First convolution	Filter 96 Channel 3 Width,Height 11 Stride 4 Zero padding 0 Output 55*55*96	Filter 96 Channel 3 Width,Height 11 Stride 4 Zero padding 0 Output 55*55*96	Filter 96 Channel 3 Width,Height 11 Stride 4 Zero padding 0 Output 55*55*96
First max pooling	Width,Height 3 Stride 2 Output 27*27*96	Width,Height 3 Stride 2 Output 27*27*96	Width,Height 3 Stride 2 Output 27*27*96
Second convolution	Filter 256 Channel 96 Width,Height 5 Stride 1 Zero padding 2 Output 27*27*256	Filter 256 Channel 96 Width,Height 5 Stride 1 Zero padding 2 Output 27*27*256	Filter 256 Channel 96 Width,Height 5 Stride 1 Zero padding 2 Output 27*27*256
Second max pooling	Width,Height 3 Stride 2 Output 13*13*256	Width,Height 3 Stride 2 Output 13*13*256	Width,Height 3 Stride 2 Output 13*13*256
Concatenate	13 * 13 * 256 * 3		
Third convolution	Filter 640, Channel 768, Width,Height 3, Stride 1, Zero padding 1, Output 13*13*640		
Fourth convolution	Filter 640, Channel 640, Width,Height 3, Stride 1, Zero padding 1, Output 13*13*640		
Fifth convolution	Filter 384, Channel 640, Width,Height 3, Stride 1, Zero padding 1, Output 13*13*384		
Fifth max pooling	Width,Height 3, Stride 2, Output 6*6*384		
First affine	Flatten 13824, Neuron 4096, Dropout 0.5, Output 4096		
Second affine	Neuron 4096, Dropout 0.5, Output 4096		
Third affine	Neuron 4096, Output 6		

4. Experimental Results

We hypothesized that accuracy increases when multimodal data are used in a multi-input format for learning. In other words, we aimed to demonstrate that high accuracy values that cannot be achieved with single-input learning can be achieved with multimodal data in a multi-input format.

4.1. Data Collection

The data was simulated using a vehicle that resembled a possible driving situation, and sounds were recorded using a portable device, assuming a driving situation. To perform bias control, we used images of men and women, young and old, and deleted 20% of the total collected data.

Six class configurations were defined for the experiment, and data were created in the following manner.

c0: Normal is when the driver looks straight ahead, and the sound is expressed as relatively quiet ambient sound.

c1: Faint is when the driver looks down and rests his head on the steering wheel, and the sound is expressed as a cracking sound.

c2: Drowsy is when the driver closes his or her eyes or yawns, and the sound is expressed as a yawning sound.

c3: Surprise is when the driver opens their eyes or mouth wide, expressing the sound as a horn sound.

c4: Anger is when the driver looks out of the side window and becomes angry, and the sound is expressed as an angry voice after the horn sound.

c5: Anxiety is reflected in the driver's tense expression, and the sound is expressed as a light horn sound, followed by an exclamation sound expressing anxiety.

The images used as the three input data were made to be the same size; namely, 227 * 227 pixels. The sound spectrograms, which have a sound length of approximately two seconds, were created using Adobe Audition software. Figure 2 displays samples of the image data used in the experiment.

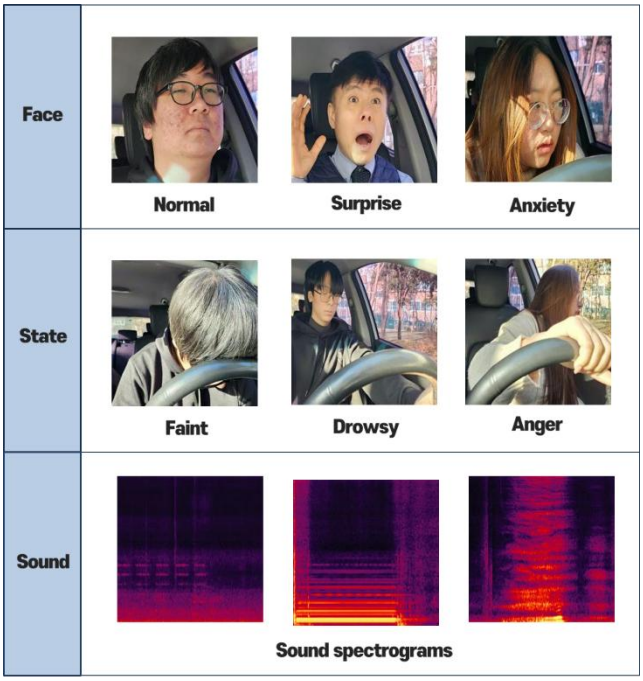


Figure 2: Examples of data

4.2. Training and Test

In total, we created 18,900 images for the experiment, 12,600 of which were used as training data and 6,300 as test data. The learning rate was 0.001, and training was performed for 200 epochs. Other experimental specifications are presented in Table 4.

Table 4: Experimental specifications

Item	Specification
Learning rate	0.001
Training epochs	200 epochs
Number of classes	6 classes
Number of data	Total data: 18,900 (6 classes * 3 inputs * 1050) Training data: 12,600 (6 * 3 * 700) Test data: 6,300 (6 * 3 * 350)
Data set	(1) Driver face images (2) Driver state images (3) Sound spectrograms

To evaluate the performance of our multi-input CNN, we conducted a parallel experiment to compare it with Resnet50 and Xception, which are well-known for their good performance. Xception consists of 14 modules and 36 convolution layers, all connected by residual connections except for the first and last convolution layers. ResNet50 has 50 layers, each trained by repeating different residual block shapes.

After training and testing using Resnet50, Xception, and our proposed multi-input CNN for single input data driver-face and driver-state, Resnet50 exhibited 81.0% and 85.7% accuracy, Xception showed 87.8% and 76.8% accuracy, and our multi-input CNN displayed 80.7% and 68.6% accuracy, respectively. Thus, all three models yielded similar results. Therefore, our proposed multi-input CNN can be considered suitable for learning prepared data. In addition, the learning result of dual input - driver-face/driver-state data - was 75.8%, which was not considerably different from the single input, but the learning result of triple input - driver-face/driver-state/sound - was 99.9%, revealing that the learning of multimodal data using the multi-input method was effective. Table 5 reports the accuracy of the learning results according to the data set for each model.

Table 5: Experimental results

Model	Data set	Accuracy (%)
Resnet50	Driver face Driver state	81.0 85.7
Xception	Driver face Driver state	87.8 76.8
Our Multi-input CNN	Driver face Driver state	80.7 68.6
	Dual input Driver face + state	75.8
	Triple input Driver face + state + sound	99.9

Figure 3 depicts the train accuracy, test accuracy and loss value for each of the four experiments using our multi-input CNN. In single-input learning for driver-face and driver-state, train and test accuracy exhibited similar trends during 200 learning epochs. However, the trend differed from single input in the case of dual input, which simultaneously inputs the driver face and state prior to learning. The training accuracy exceeded 99% from approximately 130 epochs, but the test accuracy stopped at 75% and did not increase any further. This phenomenon indicates an overfitting problem in dual input image data learning. However, in the fourth triple input experiment, where sound spectrogram data were added, the test accuracy was confirmed to have improved to 99.9%. This supports our hypothesis that accuracy increases when multimodal data are trained as multi-input.

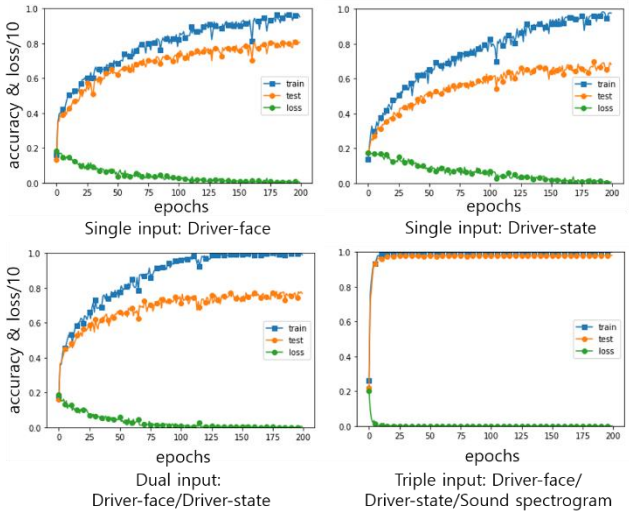


Figure 3: Test results for our multi-input CNN

5. Conclusions

This study tested the hypothesis that accuracy increases when multimodal data are trained in the form of multi-input through experiments. Three types of data were prepared: driver-face, driver-state, and sound spectrogram. In single input training for driver-face and driver-state, the test accuracy was 80.7% and 68.6%, respectively. In the case of dual input training that simultaneously inputs driver-face and driver-state, the test accuracy was 75%, but in the triple input experiment, where sound spectrogram data were added, the test accuracy improved to 99.9%. Therefore, our hypothesis that accuracy increases when multimodal data are trained in the form of multi-input was supported.

In this experiment, data were created using a digital device such as a mobile phone with a camera and recording function. We proposed a driver state recognition system with a simple multi-input CNN structure with a small number of layers that can be used for on-device AI. Before the era of

100% autonomous driving arrives, driver responsibility will continue to have an impact on traffic accidents. In future research, if body signals such as electroencephalogram (EEG), heart rate (ECG) and electrooculogram (EOG) are collected using non-contact sensors and the collected data are utilized in a multimodal manner, it is anticipated that a system with the capacity to identify more diverse driver states can be proposed.

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